

OS Architecture & Process Management - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Operating Systems | Chapter: Chapter 1: OS Architecture, System Calls & Process Management | Academic Year: 2026-2027

Dual Mode & System Calls

- * Mode Bit: 1 = User Mode; 0 = Kernel Mode.
- * Privileged instructions (I/O, halt, interrupt enable/disable) can only execute in Kernel Mode.
- * System Call sequence: User program traps to kernel -> CPU switches mode bit -> Kernel executes handler -> Returns to user mode.

Process Memory Layout

- * Stack (grows downward): Local variables, function parameters, stack frames.
- * Heap (grows upward): Dynamic memory allocated via malloc() or new.
- * Data: Initialized global/static variables. BSS: Uninitialized global/static variables.
- * Text: Read-only executable binary instructions.

Fork & Zombie / Orphan Processes

- * fork() returns 0 to child, child PID to parent, and negative on error.
- * Zombie Process: Child terminated, but parent has not yet invoked wait() to read exit status.
- * Orphan Process: Parent terminated before child; adopted by init / systemd (PID 1).